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Roll No.....

Degree: B.Tech.

Semester: SUMMER

MID-SEMESTER EXAMINATION

Course Title: *Engineering Mathematics*

Course Code: CMMTC04

Time: 1.5 Hours

Max. Marks: 25

Note: Attempt all questions. Missing data/information (if any) may be suitably assumed & mentioned in the answer.

Q. No.	Question	Marks	CO
Q 1			
1(a)	Find the inverse Laplace transform: $L^{-1} \left[\log \left(\frac{s^2 - 1}{s^2} \right) \right]$.	2.5	CO1
1(b)	Evaluate the integral $\int_0^{\infty} t e^{-3t} \sin t \, dt$ using Laplace transform.	2.5	CO1
Q 2			
2(a)	Solve the differential equation using Laplace transform $\frac{d^3 y}{dt^3} + 2 \frac{d^2 y}{dt^2} - \frac{dy}{dt} - 2y = 0$ where $y(0) = 1$, $y'(0) = 2$, and $y''(0) = 2$.	2.5	CO1
2(b)	Find the Laplace transform of $(t - 1)^2 u(t - 1)$ where $u(t - a)$ is the unit step function.	2.5	CO1
Q 3			
3(a)	Find the Laplace transform of $\frac{1 - e^{-t}}{t}$.	2.5	CO1
3(b)	Find the half range sine series of the function $f(x) = x^2$ in the interval $0 < x < l$.	2.5	CO2
Q 4			
4(a)	Find the Fourier series expansion of the function $f(x) = x \sin x$ in the interval $(-\pi, \pi)$ and deduce that: $\frac{\pi - 2}{4} = \frac{1}{3} - \frac{1}{15} + \frac{1}{35} - \frac{1}{63} + \dots$	2.5	CO2